

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA0711

COURSE OUTCOME COVERED

C05:
Measure network performance using key metrics with simulation tools
(BL 4,BL 5)

Assessment Tool Weightage: 40%

QUESTIONS: 3	Total
Marks:30	

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, wellstructured subproblems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. NETWORK CONNECTIVITY FOR A REMOTE RURAL BRANCH OFFICE

Introduction

A multinational company is establishing a new branch office in a remote rural location where high-speed wired internet services such as fiber broadband are not available. The branch needs reliable internet connectivity for cloud-based applications, video conferencing, secure communication with headquarters, email, file sharing, and other business activities.

The major constraints are limited budget, low availability of wired infrastructure, bandwidth limitations, reliability, security, and future scalability. Therefore, the network solution should provide adequate bandwidth at a reasonable cost while maintaining business continuity.

Recommended Internet Connection

The most suitable primary connection for the rural office is **4G/5G Fixed Wireless Access (FWA)**, if mobile network coverage is available.

If 4G/5G coverage is weak or unavailable, **satellite internet** can be used as the primary or backup connection. A dual-WAN arrangement can provide better reliability by using two different connectivity technologies.

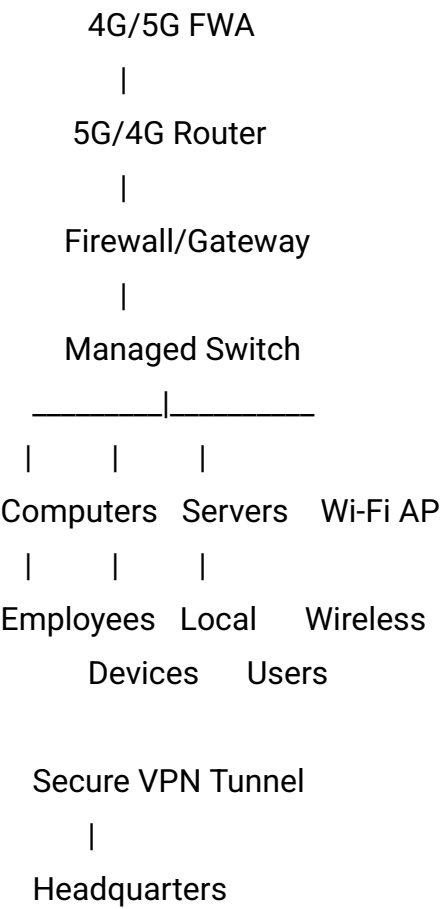
A recommended arrangement is:

Primary: 4G/5G FWA

Backup: Satellite or secondary mobile connection

This provides connectivity even though wired broadband is unavailable.

Proposed Network Architecture



Networking Devices Required

1. 4G/5G Router

A business-grade 4G/5G router should be used to receive internet connectivity from the mobile network.

Functions:

- Provides internet access through cellular networks.
- Supports multiple users.
- Can provide automatic failover.
- Supports VPN and routing features in suitable models.
- External antennas can improve connectivity in rural areas.

2. Firewall

A dedicated next-generation firewall should be installed between the internet connection and the internal network.

Functions:

- Blocks unauthorized traffic.
- Controls incoming and outgoing connections.
- Provides VPN functionality.
- Supports intrusion prevention in suitable models.
- Helps protect cloud and internal business applications.

3. Managed Network Switch

A managed Ethernet switch should connect computers, printers, access points, and other wired devices.

A managed switch is preferable because it supports:

- VLANs
- Traffic monitoring
- Access control
- Quality of Service (QoS)
- Network management

4. Wireless Access Point

Business-grade Wi-Fi access points should be deployed to provide wireless connectivity for employees.

Separate networks can be created for:

- Employees
- Guests
- IoT or other devices

5. Backup Connectivity Device

A secondary connection such as satellite or another cellular provider can be connected to the firewall using dual-WAN capability.

If the primary connection fails, the firewall can automatically switch to the backup connection.

Security Mechanisms

Security is especially important because the branch communicates with the company's

headquarters over the public internet.

VPN

A **site-to-site VPN** should be established between the rural branch and headquarters. The VPN encrypts communication and protects sensitive company data from unauthorized interception.

Firewall Security

The firewall should use:

- Stateful inspection
- Application control
- Intrusion prevention
- Web filtering where required
- Access control rules

Secure Wi-Fi

Wireless networks should use strong authentication and encryption such as **WPA2-Enterprise or WPA3-Enterprise**, depending on device compatibility.

User Authentication

Employees should use individual accounts rather than shared passwords. Multi-factor authentication should be enabled for important cloud and remote-access services.

Network Segmentation

VLANs can separate:

- Employee systems
- Guest users
- Printers/IoT devices
- Administrative systems

This reduces the impact if one device is compromised.

Bandwidth Management

Since rural wireless connectivity may have limited bandwidth, traffic should be prioritized.

Quality of Service (QoS) can prioritize important applications such as:

1. Voice and video conferencing
2. Business applications
3. VPN traffic
4. General web browsing
5. Non-business traffic

Bandwidth limits can also be applied to guest Wi-Fi so that guests do not consume the majority of the available connection.

Reliability

Uninterrupted operation is an important requirement. Therefore, the network should include:

- Primary 4G/5G connection
- Secondary backup connection
- Automatic WAN failover

- UPS for networking equipment
- External cellular antennas where required
- Network monitoring

A UPS can keep the router, firewall, switch, and access points running during short power interruptions.

Cost Considerations

The proposed solution avoids the high cost of installing new fiber infrastructure in a remote area.

The company mainly needs to invest in:

- 4G/5G router
- Firewall
- Managed switch
- Wireless access points
- VPN configuration
- Backup connection
- UPS

Operating costs consist mainly of monthly internet subscriptions and maintenance.

Future Scalability

The proposed network can be expanded easily. Additional switches and access points can be added as the number of employees increases. The firewall can support additional VPN connections, VLANs, and security policies.

The company can also migrate to fiber or another high-speed technology if it becomes available in the future.

Justification

Requirement	Proposed Solution
Wired internet unavailable	4G/5G FWA
Limited budget	Wireless connectivity instead of new fiber
Reliability	Dual-WAN and automatic failover
Security	Firewall + VPN + secure Wi-Fi
Limited bandwidth	QoS and bandwidth management
Business continuity	Backup connection + UPS
Future growth	Managed switch, VLANs and scalable firewall

Conclusion

For the rural branch office, a **4G/5G-based internet connection with satellite or secondary cellular backup** provides a practical balance between cost, bandwidth, reliability, and scalability. A firewall, managed switch, secure Wi-Fi, VPN, VLANs, QoS, and backup power should be implemented to create a secure and reliable business network. This design avoids expensive wired infrastructure while allowing the company to expand its network in the future.

2. CAMPUS-WIDE WIRELESS NETWORK FOR MORE THAN 3,000 STUDENTS

Introduction

A university wants to provide wireless internet access to more than **3,000 students** across classrooms, laboratories, libraries, hostels, and other campus areas. The network must support a large number of simultaneous users while maintaining good performance and security.

The major constraints are:

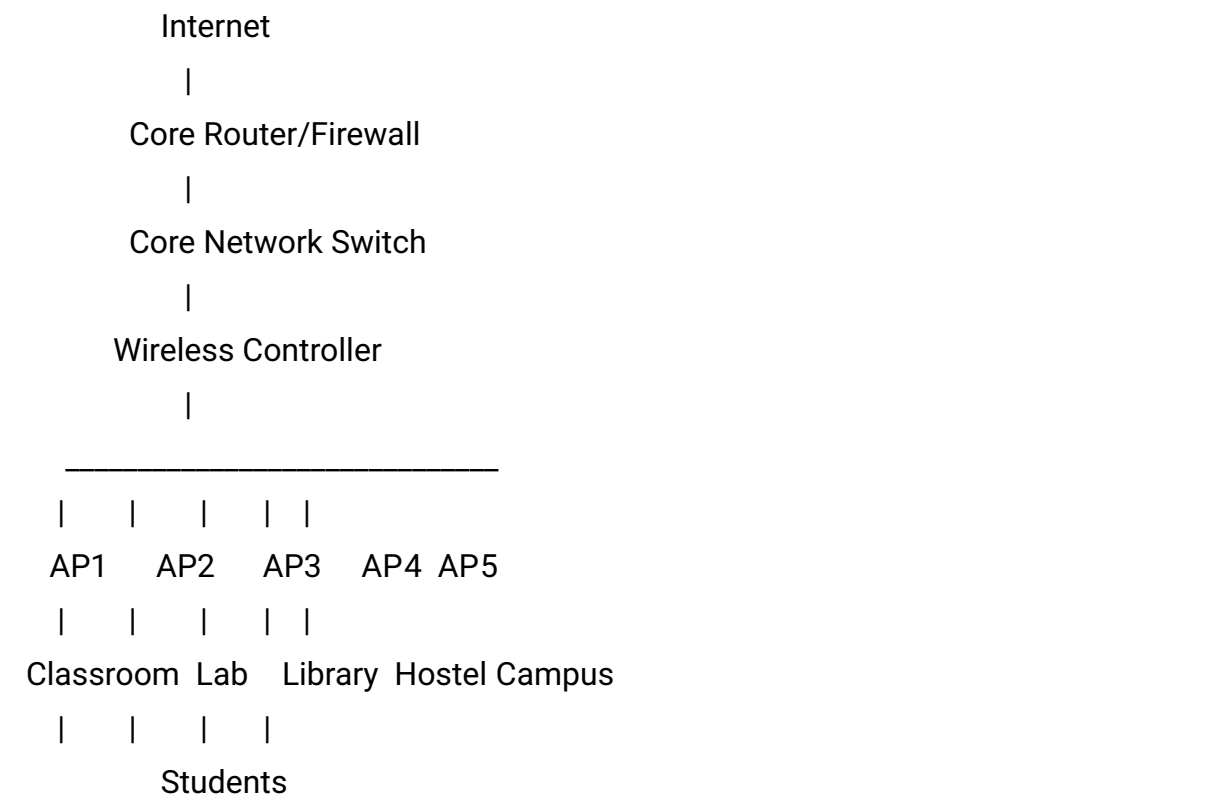
- Large number of users
- Limited number of access points
- Limited budget
- Signal interference
- High bandwidth requirements
- Secure authentication
- Wide campus coverage

Therefore, the university needs an efficient wireless design rather than simply installing access points everywhere.

Proposed Wireless Network Design

A **centrally managed Wi-Fi network** should be implemented.

The architecture can be represented as:



A wireless controller or cloud-managed system can centrally configure and monitor the access points.

Access Point Placement

Because only a limited number of APs can be deployed, their placement must be carefully planned.

1. Classrooms

Access points should be positioned to serve multiple nearby classrooms where possible.

However, placing one AP in a large classroom with many students may be necessary because student density can be more important than physical coverage.

2. Laboratories

Computer laboratories require high capacity because many computers and students may access the network simultaneously.

APs should be positioned to minimize obstruction and interference.

3. Libraries

Libraries require reliable coverage because students may use laptops, tablets, and smartphones for long periods.

High-density APs should be placed in suitable locations.

4. Hostels

Hostel buildings require careful AP placement because walls and floors can weaken Wi-Fi signals.

Access points should be distributed according to building structure and expected user density.

5. Outdoor Areas

Outdoor APs should be installed only in locations where wireless access is actually required. This helps reduce unnecessary equipment costs.

Capacity-Based Planning

Coverage alone should not determine AP placement.

For example, an access point might technically cover a large area, but if hundreds of students connect to it simultaneously, performance can become poor.

Therefore, AP placement should consider:

- Number of users
- Expected devices per student
- Application requirements
- Building construction
- User density
- Interference

The university should prioritize **high-density areas** such as lecture halls, libraries, and laboratories.

Frequency Band Selection

Modern Wi-Fi networks can use multiple frequency bands.

2.4 GHz Band

Advantages:

- Longer range
- Better penetration through walls

Disadvantages:

- More interference
- Limited number of non-overlapping channels
- Lower capacity compared with newer bands

Therefore, 2.4 GHz should mainly be used for devices that require longer range or do

not support newer bands.

5 GHz Band

The **5 GHz band** should be the primary band for most student devices.

Advantages:

- More available channels
- Less interference than 2.4 GHz in many environments
- Higher performance
- Better suited for high-density areas

6 GHz Band

If the university deploys Wi-Fi 6E/7-compatible equipment and student devices support it, the **6 GHz band** can provide additional capacity and cleaner spectrum.

However, because compatible devices and equipment can increase the cost, it should be introduced selectively.

Channel Planning

Using the same channel on nearby APs can create interference.

Therefore, the university should use a proper channel plan and automatic RF management where supported.

AP transmit power should also be adjusted rather than simply setting every AP to maximum power.

This reduces unnecessary overlap between neighboring APs.

Authentication and Security

Security is essential because thousands of students will access the network.

WPA2/WPA3-Enterprise

Enterprise-level wireless authentication should be used rather than a common password.

Students can authenticate using their university credentials.

802.1X Authentication

802.1X can be used with a RADIUS server to authenticate individual users.

Benefits include:

- Individual user accounts
- Centralized authentication
- Easier account management
- Better security
- Ability to disable individual accounts

Separate Networks

The university can create separate VLANs/SSIDs for:

- Students
- Faculty and staff
- Guests
- Administration
- IoT devices

This prevents unnecessary access between different groups.

Bandwidth Management

A large number of students can consume significant bandwidth.

The network should therefore implement:

- Per-user bandwidth limits
- QoS
- Application prioritization
- Traffic monitoring
- Fair-use policies

Academic applications should receive priority over non-essential traffic when the network is congested.

Centralized Management

A wireless controller or cloud-based management platform allows administrators to:

- Configure APs centrally
- Monitor connected users
- Detect faulty APs
- Analyze interference
- Manage firmware
- Apply security policies
- Monitor network performance

This reduces operational costs because administrators do not need to configure every AP individually.

Cost Optimization

Since the university has a limited AP budget, the design should focus on **capacity and strategic placement** .

Cost can be reduced by:

1. Deploying APs mainly in high-density areas.
2. Using centralized management.
3. Using dual-band or tri-band APs efficiently.
4. Avoiding unnecessary outdoor APs.
5. Using PoE switches so APs do not require separate power adapters.
6. Using automated RF management to optimize existing APs.

Proposed VLAN Structure

VLAN	Users	Purpose
VLAN 10	Students	Student internet access
VLAN 20	Faculty	Faculty resources
VLAN 30	Administration	Sensitive university systems
VLAN 40	Guests	Internet-only access
VLAN 50	IoT	Smart devices and equipment

Requirement Analysis

Requirement	Solution
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3,000+ students	High-density AP deployment
Limited APs	Strategic placement
Interference	Channel and power management
High performance	5 GHz/6 GHz priority
Security	WPA2/WPA3-Enterprise + 802.1X
Different user groups	VLAN segmentation
Limited budget	Capacity-based AP planning
Easy maintenance	Centralized controller

Conclusion

The university should deploy a **centrally managed, high-density wireless network** with strategically placed access points. The 5 GHz band should be the main band, while 2.4 GHz can support legacy and longer-range devices and 6 GHz can be introduced where practical. WPA2/WPA3-Enterprise with 802.1X authentication provides secure individual access. VLAN segmentation, QoS, channel planning, and centralized management improve performance and security while keeping the number of APs and overall cost under control.

3. SECURE NETWORK ARCHITECTURE FOR A FINANCIAL INSTITUTION

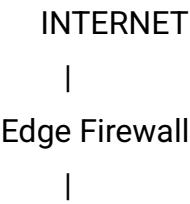
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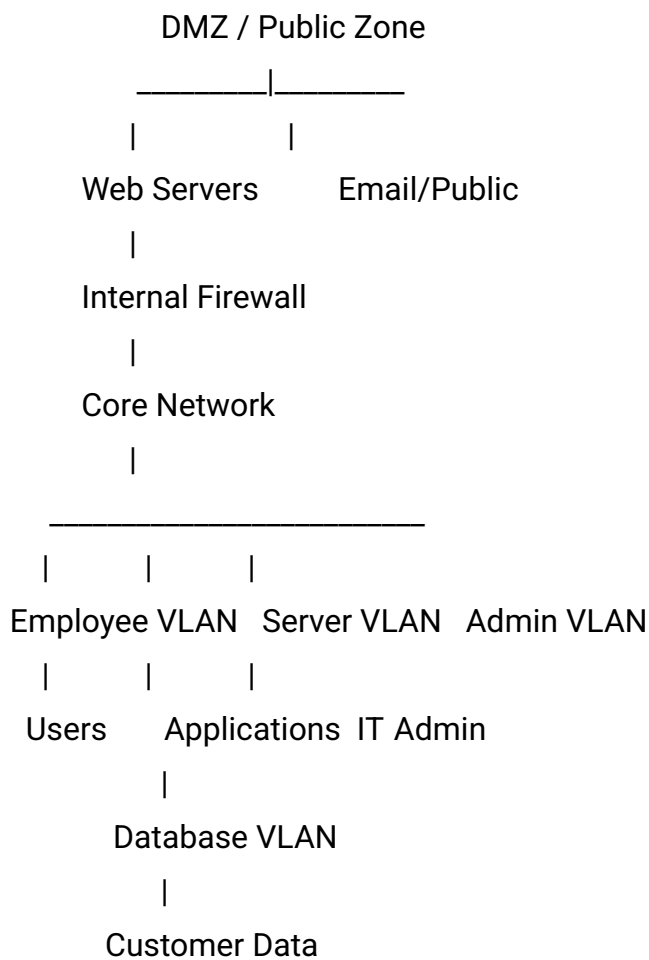
A financial institution handles highly sensitive information such as customer details, account information, transaction data, and online banking services. Therefore, cybersecurity must be a major priority when upgrading its internal network. At the same time, the organization must ensure that security controls do not significantly reduce network performance or increase operational costs.

The proposed architecture should therefore focus on:

- Strong network segmentation
- Firewall protection
- Secure authentication
- Intrusion detection and prevention
- Encryption
- Monitoring
- Compliance
- Performance
- Cost efficiency

Proposed Secure Network Architecture





A multi-zone architecture ensures that sensitive systems are separated from general employee devices.

Network Segmentation

Network segmentation is one of the most important security measures. The network should be divided into separate VLANs and security zones.

Employee VLAN

This network contains normal employee computers. Employees receive access only to the applications required for their work.

Server VLAN

Business applications and internal servers should be placed in a separate network. Employee devices should not have unrestricted access to these servers.

Database VLAN

Customer databases should be isolated from employee networks. Only authorized application servers should be allowed to communicate with database systems.

Administration VLAN

IT administrators should have a separate secure network for managing network devices and servers.

Guest VLAN

Guest users should be completely separated from internal financial systems. They should receive internet-only access.

Firewall Deployment

Firewalls should be deployed at important network boundaries.

Internet Firewall

The external firewall protects the organization from internet-based attacks.

It can provide:

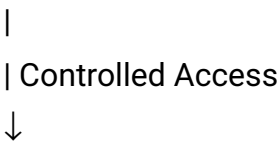
- Packet filtering
- Stateful inspection
- Application control
- Intrusion prevention
- VPN
- Web filtering

Internal Firewall

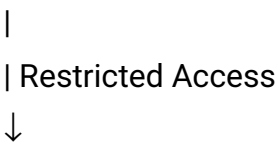
An internal firewall can control traffic between important zones.

For example:

Employee Network



Application Servers



Database Network

This ensures that even if an employee computer becomes compromised, direct access to the database is restricted.

DMZ

Public-facing systems such as web servers should be placed in a **DMZ (Demilitarized Zone)**.

The DMZ separates internet-facing systems from the internal corporate network.

For example:

Internet



Firewall



DMZ



Web/Application Server



Internal Firewall



Internal Network

This reduces the risk of an internet-facing server providing direct access to internal systems.

Authentication Methods

Strong authentication is essential for a financial institution.

Multi-Factor Authentication

MFA should be used for:

- Administrators
- Remote employees
- Access to sensitive applications
- Privileged accounts

MFA requires more than one authentication factor, reducing the risk of compromised passwords.

Role-Based Access Control

Employees should receive permissions based on their job roles.

For example:

Role	Access
Customer Service	Customer information required for work
Finance Staff	Financial systems
IT Administrator	Infrastructure management
Manager	Approved management applications
Guest	Internet only

This follows the **principle of least privilege** .

Intrusion Detection and Prevention

The institution should deploy **IDS/IPS** mechanisms.

IDS – Intrusion Detection System

IDS monitors network traffic and identifies suspicious activities.

It can generate alerts when potential attacks are detected.

IPS – Intrusion Prevention System

IPS can automatically block or prevent known malicious traffic.

A modern firewall may provide integrated intrusion prevention capabilities, reducing the need for separate hardware and helping control costs.

Encryption

Sensitive communication should use encryption.

Examples include:

- HTTPS/TLS for web applications
- Encrypted VPN connections for remote access
- Secure protocols for internal communication
- Encryption of sensitive stored data where appropriate

Encryption helps protect customer and business information from unauthorized access.

Security Monitoring

A centralized logging and monitoring system should collect security events from:

- Firewalls
- Servers
- IDS/IPS

- Network devices
- Authentication systems
- Critical applications

Security administrators can use these logs to detect suspicious activities.

A SIEM system can help correlate events from multiple sources and generate alerts.

Performance Considerations

Security should not unnecessarily slow down business operations.

The institution can maintain performance by:

1. Using high-performance firewalls.
2. Applying security inspection where it is most important.
3. Using VLANs to reduce unnecessary broadcast traffic.
4. Implementing QoS for critical applications.
5. Monitoring network utilization.
6. Avoiding unnecessary security appliances when existing firewalls provide equivalent functions.

Compliance Considerations

Financial organizations are normally subject to strict regulatory and security requirements.

The network should therefore support:

- Access control
- Audit logging
- Data protection
- Secure authentication
- Monitoring
- Incident detection
- Regular security assessments
- Proper retention of security logs

The exact compliance requirements depend on the country and services provided by the institution, so the architecture should be mapped to the applicable financial-sector regulations and organizational policies.

Cost Optimization

Security does not always require purchasing a large number of separate devices.

Costs can be controlled by:

- Using integrated next-generation firewalls
- Centralizing network management
- Using VLAN-based segmentation
- Automating monitoring and alerts
- Using existing infrastructure where secure
- Prioritizing protection for critical systems
- Regularly reviewing unused accounts and network rules

Zero Trust Approach

A **Zero Trust** approach can further improve security.

The basic principle is:
Users and devices should not automatically be trusted simply because they are inside the corporate network.
Access should be based on:

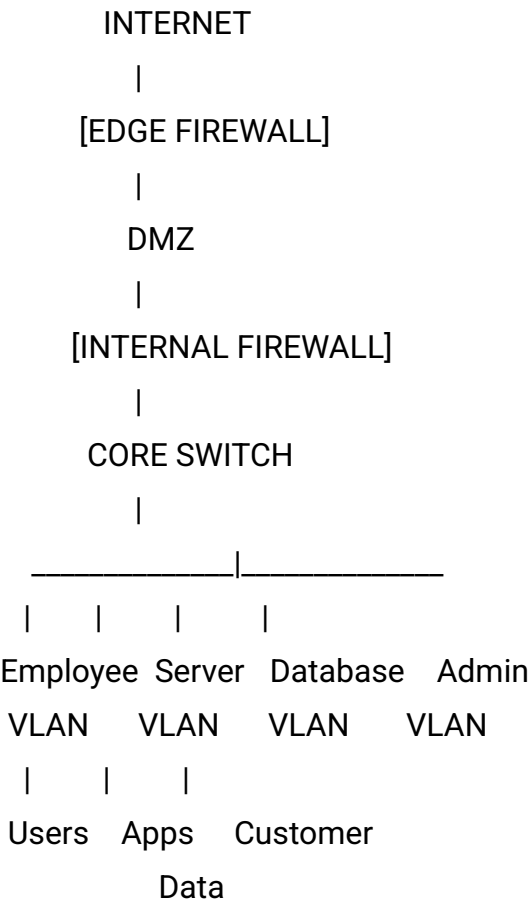
- User identity
- Device status
- Application
- Role
- Security policy
- Required resource

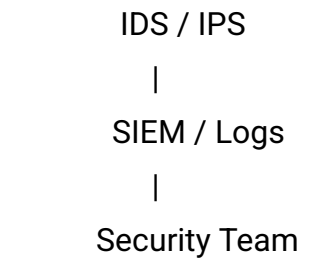
This reduces the impact of compromised accounts or devices.

Proposed Security Zones

Zone	Purpose	Security Level
Internet	External traffic	Untrusted
DMZ	Public-facing servers	High
Employee VLAN	Employee systems	Controlled
Application VLAN	Business applications	High
Database VLAN	Customer information	Very High
Admin VLAN	Infrastructure management	Very High
Guest VLAN	Visitor access	Restricted

Overall Security Flow





Requirement Analysis

Requirement	Recommended Solution
High security	Firewalls + IDS/IPS
Customer data protection	Database segmentation
Unauthorized access prevention	MFA + RBAC
Public web services	DMZ
Internal security	VLANs + internal firewall
Attack detection	IDS/IPS + centralized monitoring
Compliance	Logging, access control and auditing
Performance	High-performance firewall + QoS
Cost control	Integrated security solutions
Future scalability	Modular VLAN and security architecture

Conclusion

The financial institution should implement a **segmented, defense-in-depth network architecture**. Internet-facing services should be placed in a DMZ, while employee, application, database, administrative, and guest networks should be separated using VLANs and firewalls. Strong authentication such as MFA, role-based access control, IDS/IPS, encryption, centralized logging, and continuous monitoring provide multiple layers of protection.

This architecture balances **security, performance, compliance, and cost** by protecting the most sensitive systems with stronger controls while avoiding unnecessary restrictions on normal business activities. It is also scalable because additional VLANs, security policies, servers, and users can be added as the organization grows.

OVERALL COMPARISON OF THE THREE SOLUTIONS

Problem	Main Constraint	Recommended Solution	Main Security
Rural Branch Office	No wired broadband + limited budget	4G/5G FWA + backup connection	Firewall + VPN
University Campus	3,000+ users + limited APs	High-density centralized Wi-Fi	WPA2/WPA3-Enterprise + 802.1X
Financial	High security +	Segmented secure	Firewall + MFA +

Institution

compliance

architecture

IDS/IPS

Overall Conclusion

Constraint-based problem solving requires selecting a network design that satisfies the most important requirements while working within limitations such as **budget, bandwidth, infrastructure availability, number of users, security requirements, and scalability**.

For the rural branch, wireless broadband with backup connectivity provides an economical and reliable alternative to wired internet. For the university, strategic high-density Wi-Fi deployment with centralized management provides efficient coverage for thousands of students while controlling cost. For the financial institution, segmentation, firewalls, strong authentication, IDS/IPS, and monitoring provide layered security for sensitive financial information.

Thus, each solution demonstrates how **network performance, reliability, security, cost, and future scalability can be balanced according to the specific constraints of the organization**.